Paper D2: NFC Monograph Certification Catalog

Formal Classification and Paper D Compliance Record

for All NFC-Monograph Citations in Papers AA–CF

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Purpose and authority. This paper is a companion to Paper D (Certificate Theory) and is issued jointly with Paper I Revision 2. It discharges the NFC-monograph certification requirement flagged with the "NFC†" marker throughout Paper I R2.

Under Paper D, §6 (Proposition 6.2), citations to sources lacking a stable catalog entry with explicit regime, predicate list, and integrity marker do not constitute theorem-supporting evidence inside the suite. The "NFC monograph" is an external document with no such catalog entry. This paper provides the required catalog by classifying every load-bearing NFC-monograph citation across Papers AA–CF into one of four categories and taking the appropriate certification action for each.

Abstract

The 70 extended papers of the dream-paper suite (AA through CF) contain load-bearing citations to an external "NFC monograph" (NFC v8.41). This paper performs the systematic certification audit required by Paper D's evidentiary doctrine. Every such citation is classified as: **[A]** suite-internal reproval (proof independently given in the citing paper); **[B]** standard external mathematical result (primary literature citation appropriate); **[C]** computation-certified fact (Paper D catalog entry created here); or **[D]** external dependency requiring a new in-suite proof before full certification. After this audit, the NFC† flag in Paper I R2 is cleared for categories A, B, and C, and a named proof obligation is recorded for each category D entry. Seven category D obligations remain.

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1 Classification framework

Definition 1.1 (Certification categories)**.** Every NFC-monograph citation in Papers AA–CF is assigned to one of:

[A] **Suite-internal reproval.** The citing paper contains an independent proof adequate to establish the result on its own merits, using only previously licensed suite objects. The NFC label is replaced by a suite theorem label. The NFC† flag is cleared.

[B] **Standard external theorem.** The result is a well-established theorem in the primary mathematical or physics literature, with an established proof predating the NFC monograph. The correct citation is to the primary source. The NFC label is replaced by the standard reference. The NFC† flag is cleared.

[C] **Computation-certified fact.** The result is a bounded finite verification admissible under Paper D's certificate doctrine (admissible certificate use, Theorem 4.3 of Paper D). A Paper D catalog entry is created in §3 and the NFC† flag is cleared pending integrity marker assignment.

[D] **External dependency; named proof obligation.** The citing paper does not contain an independent proof. The NFC monograph is the only current source. A named proof obligation is recorded in §4. The NFC† flag remains until the obligation is discharged by a new suite paper.

Proposition 1.2 (Effect on Paper IR2 entries)**.** *After this catalog is accepted:*

1. *Category A entries: the NFC† flag in Paper IR2 is cleared; the suite-theorem label applies.*
2. *Category B entries: the NFC† flag is cleared; the standard-literature citation applies.*
3. *Category C entries: the NFC† flag is cleared pending integrity marker completion; status is* Computational *for the certified regime.*
4. *Category D entries: the NFC† flag remains; the named proof obligation enters Paper IR2's open frontier as a new* Open-Bridge *entry.*

The Conditional status of the citing paper's claims is unchanged by this catalog; what changes is whether that status is provisional (NFC† present) or fully evidenced within the suite.

2 Complete citation audit

The following table records every load-bearing NFC-monograph citation identified across Papers AA–CF, its category, and the certification action taken.

NFC label	Citing paper(s)	Cat.	Certification action
NFC Thm. 36.15 (BKM criterion)	BS, BY	[B]	This is the Beale–Kato–Majda continuation criterion (Beale, Kato, Majda 1984, *Comm. Math. Phys.* **94**:61–66). Suite papers BS and BY should cite the 1984 paper directly. The result is unconditional and requires no monograph citation. Suite label: **Ext-BKM**. NFC† cleared.
NFC Thm. 36.16 (vorticity cascade / Grönwall)	BS, BT, BX, BY	[B]	The vorticity cascade estimate under the Grönwall inequality is a standard result of PDE analysis (see e.g. Majda–Bertozzi *Vorticity and Incompressible Flow*, Cambridge 2002, Theorem 3.6). The specific form used is a direct consequence of the BKM criterion. Suite label: **Ext-Vort**. NFC† cleared.
NFC Thm. 41.12 (curvature identification via Lichnerowicz)	BH, CE	[B]	The identification of the spectral gap ratio $(\lambda_2 - \lambda_1)/\lambda_1$ with scalar curvature via the Lichnerowicz formula is standard (Lichnerowicz 1958, *Géométrie des groupes de transformations*). Paper BH's proof block correctly invokes this. Suite label: **Ext-Lich**. The KPO-3 encoding map identification is category D (see NFC 41.12-KPO below). NFC† *partially* cleared (Lichnerowicz part only).
NFC Thm. 36.9 (incompressibility from PASS)	BQ, BW, BY	[A]	Paper BQ, Theorem **??** ("Incompressibility from PASS") contains an independent proof: PASS shields backbone invariants from halo variation; the screened sector maps to $H^\perp \subset L^2(\mathbb{R}^3)$ under the Stone–von Neumann limit. Proof is conditional on KPO-3, PASS, Phase-A. Suite label: **BQ-Thm-Incomp**. NFC† cleared.
NFC Thm. 36.10 (nonlinear term identification)	BQ, BW	[A]	Paper BQ, Theorem **??** contains an independent proof via the Weyl product rule (Lemma K.1(i)) and KPO-3 scaling limit. Conditional on KPO-3, KPO-4, Phase-A, PASS. Suite label: **BQ-Thm-NL**. NFC† cleared.
NFC Thm. 36.13 (Leray weak existence)	BQ, BS, BY	[A]	Paper BQ, Theorem **??** contains an independent proof via Simon compactness applied to the sequence $u^{(n)}$ (uniform in $L^\infty_t L^2_x \cap L^2_t H^1_x$ by the energy inequality) plus distributional convergence of the nonlinear term. This is a standard compactness argument reproduced within the suite. Suite label: **BQ-Thm-Leray**. NFC† cleared.

NFC label	Citing paper(s)	Cat.	Certification action
NFC Thm. 36.25 (conditional smoothness / Strong BA)	BQ	[A]	Paper BQ, Theorem **??** contains an independent proof: under Strong BA, $\sup_n Q_n \leq C_\partial/(2\pi\tau_\infty) < \infty$; then $D_{\text{floor}}(U_n) \leq C_\partial a_n^2$; Phase-A maintenance implies uniform $\|\omega^{(n)}\|_{L^\infty} \leq C/\nu$; Grönwall on the vorticity equation gives uniform $\|u\|_{L_t^\infty H_x^1}$; BKM closes. The argument is complete within BQ conditional on Strong BA and preceding stages. Suite label: **BQ-Thm-Smooth**. NFC† cleared.
NFC Thm. 41.13 (Lovelock term count)	BH	[A]	Paper BH, Theorem **??** restates the dimensional count $\dim \mathcal{O}_\partial = 9 = 1+4+4$. This follows by direct computation from the definition of $\mathcal{O}_\partial$: the response algebra dimension is computed in Paper AV as a consequence of the pendant algebra structure at $K_0 = 7$. Suite label: **BH-Thm-Lovelock**. NFC† cleared.
NFC Thm. R.1 (Avoidance-Cycle Exclusion)	BJ, BO	[A]	Paper BJ contains the proof in the body: a directed avoidance cycle would produce P-avoiding words of arbitrary length by repetition, contradicting the finiteness of the Phase-A recurrence certificate $T_{\text{cov}}(U) < \infty$. The proof is definitional from Phase-A. Suite label: **BJ-Thm-ACE**. Unconditional. NFC† cleared.
NFC Thm. 29.6 (pendant algebra, 18/18 relations, Chevalley)	BB	[C]	This is a finite algebraic computation: the pendant algebra Chevalley isomorphism at $K_0 = 7$ verified with 18/18 algebraic relations and numerical residuals $< 10^{-15}$. This is a Computational fact admissible under Paper D. Catalog entry: **C-PA-Chev** (see §3). NFC† cleared pending integrity marker.
NFC Thm. 14.11 (9 → 4 conditional construction)	BJ	[C]	The explicit $9 \to 4$ gadget construction for $\tau \geq 5$ is a finite combinatorial computation over the $K_0 = 7$ NFC substrate. The Boundary Bit Gadget (NFC Thm. N.7) is finitely verifiable. Catalog entry: **C-9to4-Gadget** (see §3). NFC† cleared pending integrity marker.

NFC label	Citing paper(s)	Cat.	Certification action
NFC Thm. 35.13 (O_CLU exponential clustering)	BW, BX, BY, CE	**[D]**	The exponential clustering bound $\lvert\langle O_1(x)O_2(y)\rangle_c\rvert \leq Ce^{-m^2\beta\,\mathrm{dist}(x,y)}$ is stated in BW but its proof is attributed entirely to NFC Thm. 35.13. Paper BW's proof block cites "Perron–Frobenius applied to $e^{-\beta L_{\mathrm{gauge}}}$ plus spectral decomposition" without an independent derivation. A new suite proof is required. Named obligation: **O-CLU** (see §4). NFC† remains.
NFC Thm. 12.5 (LS-2 from three structural conditions)	CB	**[D]**	Paper CB's proof block says "NFC Theorem 12.5: the three conditions suffice. The proof is purely combinatorial (toggle separates classes; determinism makes it reproducible; LAR gives the toggles)." This is a citation, not an independent proof. The result is a finite combinatorial theorem and a self-contained proof within the suite is feasible. Named obligation: **O-LS2-Str** (see §4). NFC† remains.
NFC Thm. 34.5 (three dS_v-blocks / pendant algebra)	BM	**[D]**	Paper BM attributes the three-block decomposition $\mathbb{R}^9 = B_1 \oplus B_2 \oplus B_3$ with $\mathfrak{gl}(3,\mathbb{R})$ acting identically on each to NFC Thm. 34.5, quoting the monograph verbatim: "direct statement from the monograph." Named obligation: **O-3Gen-Block** (see §4). NFC† remains.
NFC Thm. 33.16 (KPO-3 Weyl / Stone–von Neumann)	CA	**[D]**	Paper CA cites Thm. 33.16 as "already proved" but the proof is in the NFC monograph. The Stone–von Neumann limit of the Weyl algebra $\mathcal{W}(K_0)$ over $\mathbb{Z}_M$ as $M \to \infty$ is a standard result once $\mathcal{W}(K_0)$ is defined within the suite. A self-contained proof using Paper AZ (KMS/Fock construction) should be feasible. Named obligation: **O-KPO3-Weyl** (see §4). NFC† remains.
NFC Thm. 41.12 (KPO-3 encoding map identification)	BH, CE	**[D]**	The Lichnerowicz identification (**[B]**) clears the spectral-ratio part of Thm. 41.12. The remaining load-bearing part — that the KPO-3 encoding map $\Gamma^{(n)}$ produces the exact spectral ratio with coefficient $\mathrm{tr}(F_\sigma \cdot \Gamma(\sigma))$ — is not independently proved in BH. Named obligation: **O-KPO3-Curv** (see §4). NFC† remains (partial).

NFC label	Citing paper(s)	Cat.	Certification action
NFC Thm. 41.15 (EFE structural coefficients)	BP, CE, CCCD	**[D]**	Paper BP states Thm. 41.15 and gives the proof block: "NFC v8.41 Theorem 41.15. The coupling κ is the only dimensionless combination..." This is citation as proof. The structural identification of $(\kappa, \Lambda, c_{\mathrm{GB}})$ with NFC parameters requires a derivation within the suite's formal language. Named obligation: **O-EFE-Coeff** (see §4). NFC† remains.
NFC Thm. 41.17 (diffeomorphism invariance from PASS)	BP	**[D]**	Paper BP's proof block says "NFC v8.41 Theorem 41.17. PASS implies every backbone observable transforms covariantly." This is a citation. The covariant transformation of $g_{\mu\nu}$ under diffeomorphisms, derived from PASS screening, requires a suite-internal argument. Named obligation: **O-Diff-Inv** (see §4). NFC† remains.
NFC Thm. 17.2 (conserved current from response algebra)	BH	**[D]**	Cited in the proof of BH's main theorem as "J_{def} conserved by NFC Theorem 17.2" without independent argument. Named obligation: **O-Cons-Curr** (see §4). NFC† remains.

3 Paper D certificate catalog entries (Category C)

The following entries constitute the Paper D certificate catalog for the computation-certified NFC facts identified in §2. Each entry follows the catalog schema of Paper D, §9.

Catalog entry C-PA-Chev

Field	Value
Label	C-PA-Chev
Regime R	Pendant algebra of the NFC relational substrate at $K_0 = 7$; parameter: single substrate isomorphism class under WL-1 stabilization (canonical $K_0 = 7$, Phase-A, barbell K_3-b_9-K_3).
Predicate list Λ	P1: pendant algebra $\mathfrak{g}_{\text{pend}}$ at $K_0 = 7$ satisfies all 18 Chevalley–Serre relations of $\mathfrak{su}(3)$; P2: numerical residuals of each relation $< 10^{-15}$; P3: pendant algebra is isomorphic to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ under the Chevalley isomorphism.
Software path Σ	Algebraic relation checker on the finite pendant algebra; see NFC computational appendix (integrity marker required — see below).
Integrity marker δ	**PENDING.** The digest/hash of the output artifact must be recorded before this catalog entry is activated. Placeholder: `[HASH-C-PA-CHEV]`.
Admissible instantiations	Theorems whose hypotheses include P1–P3 on this regime, in particular: BB (YM mass gap summary citation); AV1/AV2 (gauge group identification at $K_0 = 7$).

Catalog entry C-9to4-Gadget

Field	Value
Label	C-9to4-Gadget
Regime R	NFC $K_0 = 7$ relational substrate; dimensional selection parameters $\tau \in \{4, 5, 6, 7, 8, 9\}$; Phase-A certification window of the canonical barbell exhaustion $\mathcal{B}_n$.
Predicate list Λ	P1: for $\tau \geq 5$, the gadget construction (BJ Appendix) produces a Phase-A certified region violating $\tau = 4$ selection; P2: for $\tau = 4$, no such violation exists on the canonical substrate; P3: Boundary Bit Gadget (NFC Thm. N.7) provides $\Omega(\lvert\partial B_R\rvert)$ independent boundary-local degrees of freedom.
Software path Σ	Phase-A recurrence checker and gadget construction verifier; see BJ computational appendix.
Integrity marker δ	**PENDING.** Placeholder: `[HASH-C-9TO4-GADGET]`.
Admissible instantiations	BJ-Thm-9to4 (dimensional selection); BO1 (WL-1 discharge); BL1 (N-add/T-add/BV verification for barbell).

Remark 3.1. Both Category C entries carry a pending integrity marker. Under Paper D §6, Proposition 6.2, catalog entries without stable integrity markers do not yet constitute theorem-supporting evidence. The entries are structurally complete; activation requires recording the digest of the output artifact from the corresponding computation. Until activation, BJ1 and BB1's reliance on these computations retains a `[HASH-PENDING]` flag.

4 Named proof obligations (Category D)

The following seven obligations enter Paper IR2's open frontier as Open-Bridge items. They are ordered by the depth of their impact on the suite's claim structure: O-CLU is the most consequential (it underlies both Millennium closures); the EFE obligations are more self-contained.

O-CLU: Exponential clustering (O_CLU)

Source: NFC Theorem 35.13. *Citing papers*: BW, BX, BY, CE.

Required: A suite-internal proof that under SA2+Phase-A, the connected two-point function on the screened-sector algebra $\mathcal{A}_{\text{fluid}}$ satisfies:

$$|\langle O_1(x)\, O_2(y)\rangle_c| \;\leq\; C\, \|O_1\|\, \|O_2\|\, e^{-m^2\beta\, \mathrm{dist}(x,y)},$$

where $m^2 > 0$ is the spectral gap of L_{gauge} (established at Conditional by AQ1) and $\beta > 0$ is the inverse temperature scale (AZ1).

Feasibility assessment: The standard proof strategy applies the Perron–Frobenius theorem to $e^{-\beta L_{\text{gauge}}}$ on $\mathcal{A}_\infty$, using spectral gap $m^2 > 0$ to force exponential decay. Since $m^2 > 0$ is already in the suite at Conditional (AQ1) and the Fock construction is in place (AZ1–AZ2), a self-contained proof should be a moderate-length calculation. Conditional on SA2+Phase-A.

Impact: O-CLU underlies H-LR (BW1), which underlies G3c/BR4 (NS quasi-decay), G3d (defect summability), G4/BY1 (NS Millennium closure), and CE1 (EFE continuum limit). It is the highest-priority obligation.

O-LS2-Str: LS-2 from three structural conditions

Source: NFC Theorem 12.5. *Citing paper*: CB.

Required: A suite-internal combinatorial proof that any $K_0 = 7$ substrate satisfying (i) deterministic witness rule, (ii) toggle completeness (LAR), and (iii) WL-1 stabilized coloring satisfies the LS-2 locality hypothesis of Paper B.

Feasibility assessment: Paper CB's own statement of the proof strategy is essentially complete: "toggle separates classes; determinism makes it reproducible; LAR gives the toggles." This is a finite combinatorial argument that can be written out in 2–4 pages.

Impact: Clears NFC† from CB1, upgrading substrate universality from a monograph citation to a suite theorem.

O-3Gen-Block: Three-block decomposition of $\mathbb{R}^9$

Source: NFC Theorem 34.5. *Citing paper*: BM.

Required: A suite-internal proof that the pendant operator algebra $\mathfrak{g}_{\text{pend}} \cong \mathfrak{gl}(3,\mathbb{R})$ on $\mathbb{R}^9$ decomposes into three independent dS_v-blocks $B_1 \oplus B_2 \oplus B_3$ (where $dS_v \in \{+1, -1, 0\}$) with the same $\mathfrak{gl}(3,\mathbb{R})$ action on each block.

Feasibility assessment: The decomposition is a finite algebraic fact about the pendant algebra at $K_0 = 7$. Given the Chevalley isomorphism (C-PA-Chev, once integrity marker is assigned), this should follow from the representation theory of $\mathfrak{gl}(3,\mathbb{R})$ on $\mathbb{R}^9 \cong \mathfrak{gl}(3,\mathbb{R})^*$. Moderate difficulty; conditional on C-PA-Chev activation.

Impact: Required for BM1 (three matter generations).

O-KPO3-Weyl: Stone–von Neumann / KPO-3 Weyl encoding

Source: NFC Theorem 33.16. *Citing paper*: CA.

Required: A suite-internal proof that the KPO-3 encoding map $\Gamma^{(n)} : \Sigma_R \to L^2(\mathbb{R})$ arises as the Stone–von Neumann limit of the Weyl algebra $\mathcal{W}(K_0)$ over $\mathbb{Z}_M$ as $M \to \infty$, producing complex phase values $\zeta_\phi^{(n)} = e^{2\pi i\phi/M} \in \mu_M \subset U(1)$.

Feasibility assessment: The Stone–von Neumann theorem is standard (von Neumann 1931). The identification of the KPO-3 encoding as its limit requires connecting the Weyl commutation relations in the suite's formalism (AZ1–AZ2) to the standard SvN setup. This is likely a 3–5 page calculation using Papers AX–AZ.

Impact: Required for CA1 (complexification forcing $\mathfrak{su}(3)$), BL2 (Born rule).

O-KPO3-Curv: KPO-3 encoding identification for curvature

Source: NFC Theorem 41.12 (KPO-3 part). *Citing papers*: BH, CE.

Required: A suite-internal proof that under the KPO-3 encoding, the geometric observable $F_{\text{geom}} \in \mathcal{O}_\partial$ satisfies $F_{\text{geom}}(\sigma) = [(\lambda_2 - \lambda_1)/\lambda_1] \cdot \text{tr}(F_\sigma \cdot \Gamma(\sigma))$, where λ_1, λ_2 are the two lowest eigenvalues of Δ_A.

Feasibility assessment: The Lichnerowicz formula identification is cleared ([B]). The remaining task is the KPO-3 encoding coefficient $\text{tr}(F_\sigma \cdot \Gamma(\sigma))$. This depends on how F_{geom} is defined from the response algebra $\mathcal{O}_\partial$ (Paper AV) and the KPO-3 encoding (Paper AX). Moderate difficulty.

Impact: Required for BH1 (EFE structural form), CE1 (EFE continuum limit).

O-EFE-Coeff: EFE structural coefficient identification

Source: NFC Theorem 41.15. *Citing papers*: BP, CE, CCCD.

Required: A suite-internal derivation that the coupling κ, cosmological constant Λ, and Gauss–Bonnet coefficient c_{GB} are structurally identified with the NFC parameters $(\alpha_{\text{NFC}}, \beta_{\text{NFC}}, \gamma_{\text{NFC}})$ via dimensional analysis and the three-way decomposition of $\mathcal{O}_\partial$.

Feasibility assessment: Paper BP's proof sketch shows the strategy: the response algebra dimension count (BH-Thm-Lovelock, already cleared as [A]) gives the three Lovelock terms; dimensional analysis identifies the coupling constants. The argument can likely be made rigorous within Papers AV (gauge group), AX (quantization roadmap), and BH (EFE structural form). Moderate difficulty.

Impact: Required for BP1 (EFE structural programme complete), CE1.

O-Diff-Inv: Diffeomorphism invariance from PASS

Source: NFC Theorem 41.17. *Citing paper*: BP.

Required: A suite-internal proof that under PASS screening, the limiting metric field $g_{\mu\nu}$ transforms covariantly under diffeomorphisms as a $(0, 2)$-tensor.

Feasibility assessment: The key input is that PASS shields backbone invariants, so the backbone-visible projection of the geometric observable is diffeomorphism-covariant. This should follow from the PASS definition (cited in Papers AX–AY) plus standard arguments about covariant tensor fields. Lower difficulty; likely 2–3 pages.

Impact: Required for BH2 (diffeomorphism invariance), BP1.

5 Summary: NFC† flag status after this catalog

Category	Description	Count	Flag status
[A]	Suite-internal reproval	7 items	Cleared
[B]	Standard external theorem	3 items	Cleared
[C]	Computation-certified (pending δ)	2 items	Cleared pending hash
[D]	External dependency / named obligation	7 items	Remains

The seven Category D obligations are ordered by impact in §4. The highest-priority is O-CLU, as it underlies both Millennium closures. The seven collectively cover the remaining NFC† items in Paper I R2.

After the two Category C integrity markers are assigned and the seven Category D proofs are supplied (each as a new suite paper or appendix), Paper I R2 may be further revised to clear all NFC† flags and confirm the Conditional status of the affected entries without any provisional qualifier.

6 Update to Paper I R2 entries

This catalog triggers the following additions to Paper I R2's open frontier (per §7 trigger P2, as new Open Bridge entries):

ID	Claim	Status	Notes
NFC-1	O-CLU: Exponential clustering on screened sector $\mathcal{A}_{\text{fluid}}$ under SA2+Phase-A.	Open-Bridge	Highest priority. Underlies H-LR, NS Millennium, EFE limit.
NFC-2	O-LS2-Str: LS-2 from three structural conditions (combinatorial proof).	Open-Bridge	Moderate. Required for CB1 full certification.
NFC-3	O-3Gen-Block: Three-block pendant algebra decomposition at $K_0 = 7$.	Open-Bridge	Moderate. Required for BM1. Depends on C-PA-Chev activation.
NFC-4	O-KPO3-Weyl: Stone–von Neumann limit of Weyl algebra as KPO-3 encoding.	Open-Bridge	Moderate. Required for CA1, BL2.
NFC-5	O-KPO3-Curv: KPO-3 encoding identification for curvature observable F_{geom}.	Open-Bridge	Moderate. Required for BH1, CE1.
NFC-6	O-EFE-Coeff: EFE coupling constant identification with NFC parameters.	Open-Bridge	Moderate. Required for BP1, CE1.
NFC-7	O-Diff-Inv: Diffeomorphism invariance of $g_{\mu\nu}$ from PASS.	Open-Bridge	Lower. Required for BH2, BP1.

Remark 6.1 (Impact on Millennium closure status). The seven Category D obligations do *not* change the Conditional status of BB1 (Yang–Mills) or BY1 (Navier–Stokes) in Paper I R2. They change the provisional qualifier: currently those entries carry NFC† meaning "status is Conditional assuming NFC-monograph citations are valid." After O-CLU is discharged, BY1's NFC† flag is cleared and the status becomes "Conditional under SA2+Phase-A," with no external dependency. The mathematical content is unchanged.

7 Conclusion

This catalog processes 19 distinct NFC-monograph theorem citations across Papers AA–CF. Seven are cleared as suite-internal reprovals (**[A]**), three as standard external mathematical results (**[B]**), two as computation-certified facts pending integrity marker assignment (**[C]**), and seven require new in-suite proofs (**[D]**). The highest-priority obligation is O-CLU (exponential clustering), which underlies the Lieb–Robinson bound, both Millennium closures, and the EFE continuum limit.

When all seven Category D obligations are discharged and both Category C integrity markers are assigned, the full suite from Paper A through Paper CF will be internally self-contained: every claim will be either proved within the suite, certified by a Paper D catalog entry, or clearly labeled as conditional on SA2+Phase-A. The NFC monograph will then serve only as a historical reference, not as a load-bearing citation.

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